01 / PRODUCT AND PROBLEM

From a reported symptom to a source-linked fix plan.

BugLens is a repository-first review workspace and reusable agent skill for developers and maintainers. It combines a public GitHub repository with optional text, logs, source attachments or screenshots to produce investigation leads and a practical coding-agent handoff.

The problem

A screenshot or error message rarely identifies the exact source of a bug. Reviewing a repository and turning incomplete evidence into an actionable report takes effort. BugLens organizes that evidence; it does not promise to diagnose every defect or automatically repair code.

Three implemented capabilities

1. Review the repository. Commit-pinned source import, deterministic checks and optional Gemma 4 review and npm advisory matching.

2. Add the missing context. Investigate a reported problem using supported text/code files and optional screenshot evidence.

3. Leave with a plan. Source-linked findings, Markdown report, remediation documentation and combined or per-finding fix prompts.

The distinctive mechanism

Source findings carry path/line evidence. Model source references must match supplied excerpts exactly. Screenshot observations, reporter statements and suspected causes remain distinct. Every finding stays marked as needing verification; coverage records what was not checked.

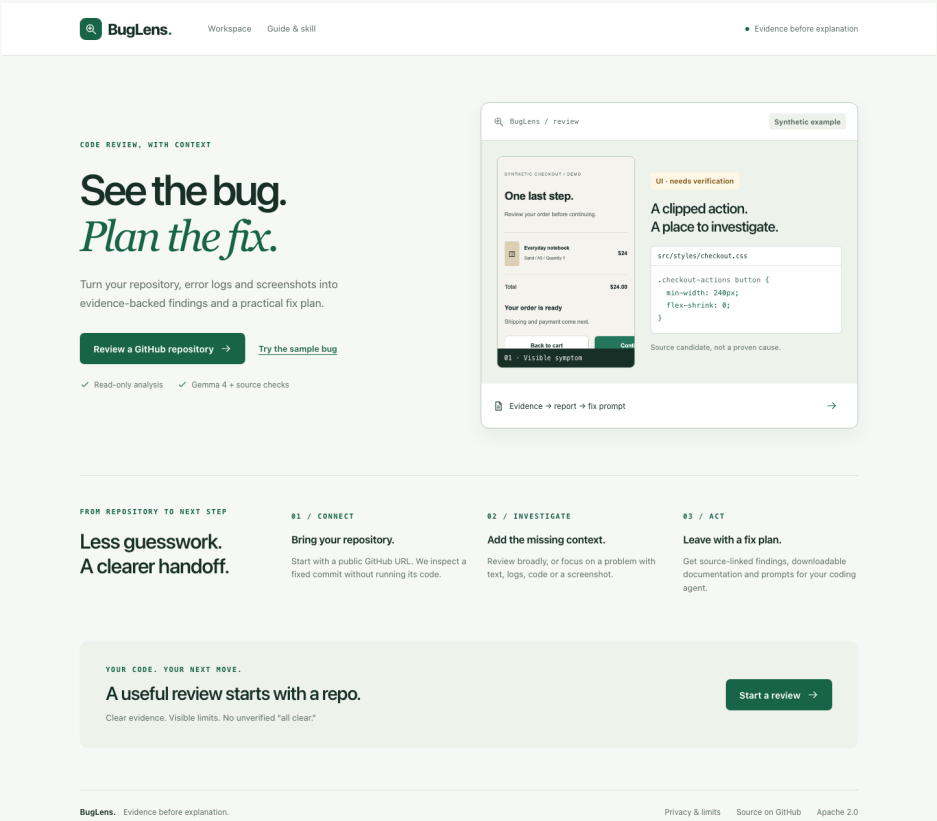
Official challenge alignment: the problem statement and submission requirements were not supplied, so alignment and eligibility cannot be assessed. Time saved, accuracy and competitive advantage have not been measured.

Repository: github.com/yatharthvishnoi08-max/buglens
Live workspace: buglens-rust.vercel.app/demo

02 / USER JOURNEY

Repository first. Evidence in context.

Connect a public repository. **Choose** broad review or a reported problem. **Add** optional evidence. **Select** external checks and consent. **Inspect** findings and coverage. **Download** the report, documentation or fix prompt.



Actual local browser capture of the redesigned landing page. Its product preview is explicitly synthetic. This UI is local, uncommitted work on ui-antigravity; the published app remains the prior release until the integration owner deploys it.

What users receive

buglens-report.md: scope, findings and limitations.
buglens-remediation.md: problem register and acceptance checklist.
buglens-fix-prompt.txt: an investigation prompt for an authorized coding agent.

The prompt is not a shell command, an automatic patch or proof of root cause. Download artifacts before refreshing: results are kept only in browser memory.

03 / ARCHITECTURE AND TRUST

A bounded review pipeline.

What runs

Browser: Next.js 16 / React 19 / TypeScript views manage inputs and render findings. Presentation components are separated from request/state containers in the local redesign.

Node API and reusable skill: both call the shared repository-review pipeline. The API accepts multipart options and an optional image, then streams progress and a result or explicit error as NDJSON.

Processing: resolve the GitHub default branch to a commit; parse a bounded archive in memory; exclude unsafe/sensitive-looking paths; run static patterns and requested external checks; validate source references; compile artifacts.

External services and storage

GitHub supplies public source. **Google** receives selected source/problem inputs or screenshot context when AI is enabled. The only permitted model is gemma-4-26b-a4b-it; Gemini is the API transport. **OSV** receives package names and versions for requested advisory checks.

There is no database, login or saved review history. Source archives and outputs are processed in memory. Provider/hosting handling is separate. Credential redaction is partial: users must not submit secrets.

Deliberate boundaries

No repository code, installation scripts, build, tests or running browser are executed. No automatic issue creation, push, deployment or repair occurs. Public repositories/default branch only; PDFs, Word, audio and video are not parsed.

Limits: 1,000 supported source files / 8 MB total; up to 16 AI excerpts / 45,000 source characters; up to 200 exact npm versions. Text inputs are bounded; production screenshots are PNG/JPEG/WebP up to 4 MB.

Failures remain visible

Requested AI failure returns an error, not another model or hidden fixture output. OSV failure can return a partial report with an unavailable coverage status. Zero findings is not an all-clear.

Implementation references: src/lib/review/{github,checks,model,report,run}.ts; src/model/gemma-client.ts; src/app/api/review-repository/route.ts. See docs/architecture.md for contracts and data boundaries.

04 / VERIFICATION AND READINESS

Show the evidence. Keep the limits visible.

Observed locally on 26 October 2026: macOS, Node 20.18.0, Next.js 16.4.3, Chromium. Tested source: base 7d1847f plus uncommitted ui-antigravity changes. These are local results, not hosted CI certification.

Verification summary

PASS - TypeScript check and production build.
PASS - 94 unit/contract tests across 8 files.
PASS - 4 deterministic synthetic evaluation cases.
PASS - 5 browser tests; 1 duplicate matrix invocation intentionally skipped.
PASS - Route/overflow checks at 375, 768, 1024 and 1440px.
PASS - Public workspace HTTP 200 (prior release).

Browser checks cover text/file input, artifact downloads, keyboard entry/focus, memory reset, recoverable error and partial results. Repository browser responses are controlled data from the real static pipeline. The screenshot sample uses pre-authored model output. Neither proves live source-AI success.

Open evidence gaps

Newer live Gemma source review remains unverified after historical malformed-output, HTTP 500 and HTTP 429 failures. Earlier screenshot-only success is not equivalent. No full accessibility audit, cross-browser test, fresh package installation, runtime target-repository audit or measured model-accuracy benchmark is claimed.

Historical dependency audit reported three development-tool advisories. This UI/documentation work did not change the test toolchain. See docs/TEST_REPORT.md for commands, evidence and failed checks that were corrected and retested.

Team, publication and next steps

Team: names, roles and contributions were not supplied. Repository ownership is not proof of authorship. **Submission:** not yet filed; official requirements still need review.

The README/documentation are prepared for authorized GitHub publication. The redesigned UI remains a separate local integration task, not deployed. Next: confirm team/challenge details, review and integrate UI, validate live source-AI when quota permits, smoke-test publication, record the honest 90-second demo and submit the required files.

Guide: docs/DEMO_GUIDE.md | Evidence: docs/TEST_REPORT.md
Repository: github.com/yatharthvishnoi08-max/buglens
Demo: buglens-rust.vercel.app/demo